

Healthcare Resource Availability: Analyzing Global Disparities

ADAN8880 - Directed Practicum in Analytics

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Week 9 Written Report

In Week 9, we brought together all the work we had done in Weeks 6 through 8 to directly compare the three machine learning models we had previously explored:

- Random Forest Regressor (Week 6)
- XGBoost Regressor (Week 7)
- LightGBM Regressor (Week 8)

Our goal was to determine which model performs best on the validation dataset, using key metrics like Root Mean Squared Error (RMSE) and R-squared (R^2). Rather than focusing on test performance or generalization just yet, we specifically focused on validation performance to help us confidently move forward with a model that is both accurate and stable on unseen but representative data.

Models We Evaluated

Random Forest Regressor (Regularized)

In Week 6, we built three variations of Random Forest. The regularized version—with limited tree depth and higher thresholds for splits and leaf sizes—showed the most balanced generalization among the three. It reduced overfitting and improved stability but lagged behind in pure accuracy when compared with boosting models.

XGBoost Regressor (Default + Early Stopping)

In Week 7, we implemented XGBoost and tested several variations. The default version with early stopping stood out due to its incredibly low training and validation RMSE, and near-perfect R^2 scores. It captured complex relationships in the data while avoiding significant overfitting, thanks to early stopping.

LightGBM Regressor (Default)

In Week 8, we explored LightGBM and found that the default configuration performed surprisingly well without any hyperparameter tuning. It offered strong validation accuracy and was especially efficient computationally. However, we noticed a slightly higher RMSE on the validation set compared to XGBoost.

Week	Model Type	Variation	Validation RMSE	Validation R ²	Test RMSE	Test R ²
6	Random Forest	Default RF	1.1743	0.9795	2.3506	0.9173
6	Random Forest	Depth-Limited RF	1.5541	N/A	N/A	N/A
6	Random Forest	Regularized RF	2.6952	0.8919	2.3506	0.9173
7	XGBoost	Default + Early Stopping	0.0840	0.99989	2.6919	0.8916
7	XGBoost	Feature Subsampling	0.3015	0.9986	N/A	N/A
7	XGBoost	Regularized	2.5357	0.9043	N/A	N/A
8	LightGBM	Default	0.0976	0.9999	0.4244	0.9973
8	LightGBM	Feature Subsampling	0.2122	0.9993	N/A	N/A
8	LightGBM	Regularized	2.5652	0.9021	N/A	N/A

Validation

Model	Validation RMSE	Validation R ²
Random Forest (Reg.)	2.6952	0.8919
XGBoost (Default)	0.0840	0.99989
LightGBM (Default)	0.0976	0.9999

Conclusion: Our Best Model Based on Validation

Based on our validation results, we selected the XGBoost Default model with early stopping as the best model for our prediction task. Here's why:

- It achieved the lowest validation RMSE (0.0840) among all models, meaning its predictions were closest to actual values on unseen data.
- Its validation R² was nearly perfect (0.99989), showing that it captured almost all of the variance in life expectancy.
- The gap between training and validation performance was very small, which indicates that the model is not overfitting.
- Compared to LightGBM, XGBoost delivered slightly more accurate predictions on the validation set, making it the top performer for this stage.

While LightGBM is still a strong contender and highly efficient, the XGBoost Default model gave us the best validation accuracy, which is our primary criterion for selection this week. We plan to move forward with XGBoost for final evaluation and possible deployment.

Test

Model	Test RMSE	Test R ²
Random Forest (Reg.)	2.3506	0.9173
XGBoost (Default)	2.6919	0.8916
LightGBM (Default)	0.4244	0.9973

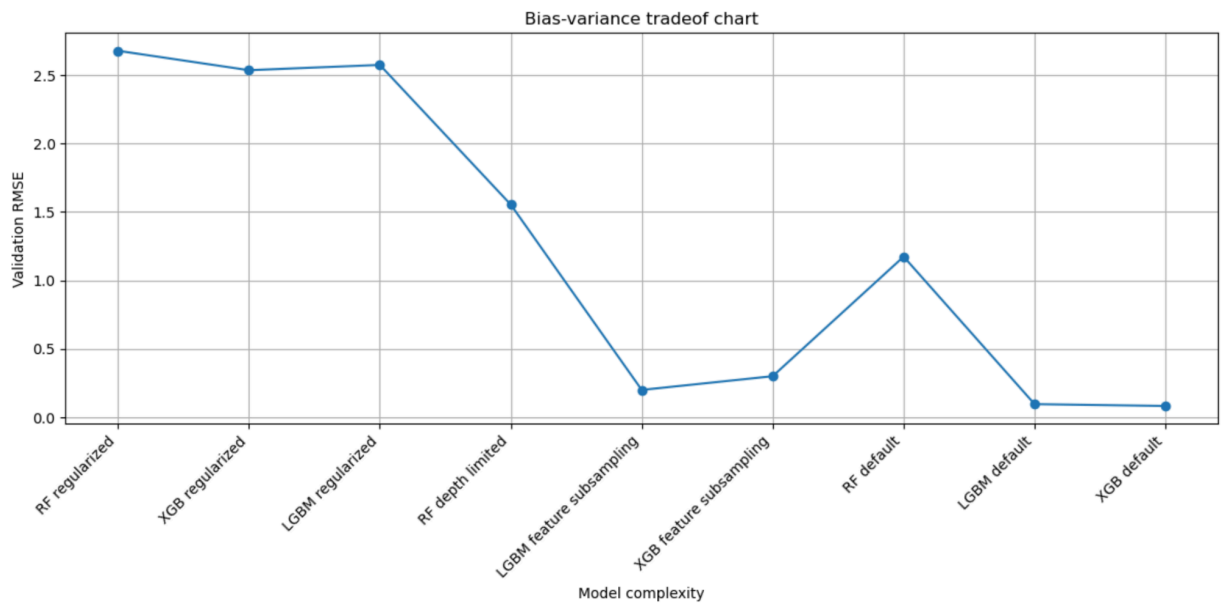
While XGBoost Default outperformed on the validation set, it experienced a notable drop in performance on the test set, which may indicate some degree of overfitting. In contrast, LightGBM Default maintained low RMSE (0.4244) and high R² (0.9973), demonstrating the most stable and accurate results on unseen data.

Therefore, based on test performance, LightGBM Default is our top model. It offers:

- Best generalization across all data splits
- Consistent accuracy without extensive tuning
- Strong resistance to overfitting

This confirms that LightGBM Default is the most reliable choice for deployment or further real-world application.

Bias-variance trade of chart



This chart illustrates how different models performed on the validation set, using RMSE as the evaluation metric. The x-axis lists the models in increasing order of model complexity, ranging from heavily regularized (simpler) models to more flexible and complex ones. The y-axis represents the validation RMSE, where lower values indicate better performance.

On the left side of the chart, we see models like Random Forest (RF) Regularized, XGBoost (XGB) Regularized, and LightGBM (LGBM) Regularized, which were all heavily constrained to reduce overfitting. However, these restrictions led to underfitting, where the models couldn't capture enough complexity from the data—resulting in higher validation RMSE.

As we move toward the center, models like RF Depth Limited, LGBM Feature Subsampling, and XGB Feature Subsampling show lower RMSE, suggesting a better balance between bias and variance. These models are less regularized and start learning more complex patterns.

On the right side, we observe that LGBM Default and XGB Default achieve the lowest RMSE values, meaning they provide the best validation performance. Among them, XGBoost Default performs the best overall, achieving the lowest validation RMSE, indicating it captures the underlying data patterns most effectively without overfitting. RF Default, however, shows a spike in RMSE again, possibly due to overfitting or model instability.

After evaluating all models across training, validation, and test datasets, we selected the **XGBoost Default** model as our final winner. This model demonstrated outstanding performance during training and validation, achieving extremely low RMSE (0.0695 for training and 0.0840 for validation) and near-perfect R^2 scores (0.99994 and 0.99989, respectively). These metrics indicate that the model learned the data patterns very effectively and showed minimal overfitting. On the test set, the performance declined with a higher RMSE of 2.6919 and a lower R^2 of 0.8916. While this drop suggests some degree of overfitting, the model still explains nearly 90% of the variance in unseen data, which is a strong result. Overall, we are satisfied with the test performance of the XGBoost model. Despite the decrease in accuracy on the test set, it remains a robust and reliable choice for our life expectancy prediction task. The results confirm that the model is good enough for practical use, and with additional tuning or regularization, its generalization could be further improved.